



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-00 Environment Setup

Windows + WSL2 beginner route | install each tool only at first use

Artifact	C3_006_C3_00_ENVIRONMENT_SETUP_BEGINNER_FIRST_RC1
Status	2026 - LEARNER EDITION
Teaching standard	M01 beginner-first teacher loop; book remains sufficient without video.
Bridge boundary	Course 2B competencies may be assumed; all new engineering concepts still start from first principles.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: understand -> follow once -> see expected result -> break/fix -> do a changed version -> validate -> explain -> save evidence. A running command alone never counts as mastery.



Environment principle

This phase uses the LIGHT profile. Start with Windows + WSL2. Git and Python are installed only when their first-use lessons reach them; Docker is delayed until L11. Save actual command output. Never fabricate a green check when a tool is unavailable.

L02 - WSL FIRST USE

Windows PowerShell: run `wsl -l -v`. If Ubuntu or another intended distribution is already VERSION 2, continue. If WSL is absent, Administrator PowerShell: `wsl --install`; restart if requested; finish Linux username setup; rerun `wsl -l -v`.

```
wsl -l -v
# Expected shape (distribution name may differ):
#   NAME      STATE      VERSION
# * Ubuntu    Running      2
```

Inside WSL create the project under `~/course3/...`. For Linux-heavy Git/Python work, do not default to `/mnt/c/...`; Microsoft currently recommends the WSL filesystem for Linux-tool performance.

L04 - GIT FIRST USE

```
git --version
# If command is missing inside WSL:
sudo apt update
sudo apt install -y git
git --version
```

Configure identity only with your real Git name/email if needed; this course does not invent credentials.

L05 - PYTHON / VENV FIRST USE

```
python3 --version
# If missing:
sudo apt update
sudo apt install -y python3 python3-venv python3-pip
python3 --version
python3 -m venv .venv
source .venv/bin/activate
python -c "import sys; print(sys.executable)"
```

Expected: active interpreter path is inside your project `.venv`. Add `.venv/` to `.gitignore`.

L05/L09 - PYTEST FIRST USE

```
python -m pip install pytest
pytest --version
pytest -q
```

Expected: pytest version prints and your current tests run. If command is missing, first verify you activated the correct venv.



L11 - Docker first use

Do not install Docker earlier. At L11 run `docker version`. If both Client and Server are present, keep the working setup. If Docker is absent, install Docker Desktop for Windows from the official instructions and use the WSL2 backend where supported.

```
docker version
# Real verification must show Client and Server information.
docker run --rm hello-world
```

If the engine cannot run safely on your laptop because of hardware, policy or virtualization constraints, record the exact blocker. Complete the approved design-only Docker fallback. Do not invent screenshots/output.

RECOVERY ORDER

1) Identify which shell you are in. 2) `pwd`. 3) verify tool version. 4) verify active Python interpreter if relevant. 5) reproduce smallest command. 6) save exact error. 7) change one thing at a time. Reinstallation is not step 1.

CURRENT AUTHORITIES

[Microsoft WSL install](#)

[Microsoft WSL environment setup](#)

[Git status](#)

[Git switch](#)

[Python 3.14 venv](#)

[pytest reference](#)

[Docker Desktop Windows install](#)